

LeetCode 2910: Minimum Number of Groups

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| Algorithmic Logic

Har group ka size sirf k ya $k+1$ ho skta hai.
k ko minimum frequency se 1 tak try karo.

Sabse pehle jitna ho sake bade group ($k+1$) banao, kyunki groups kam banenge.

```
groups = ceil((double)freq / (k + 1));
```

Ab check karo ki ye possible bhi hai ya nahi.

```
if(groups * k > freq)
```

Matlab agar groups bana diye aur har group me minimum k elements chahiye, to total $groups * k$ elements chahiye.
Agar itne elements hi present nahi hain (freq), to ye k impossible hai.

```
flag = false; break;
```

Agar valid hai to:

```
curr += groups;
```

End me:

```
if(flag)
    ans = min(ans, curr);
```

| C++ Implementation

```
class Solution {
public:
    int minGroupsForValidAssignment(vector<int>& balls) {

        int n=balls.size();
        unordered_map<int,int> mp;
        for(int ball:balls){
            mp[ball]++;
        }
        int minu=INT_MAX;
        for(auto &it:mp){
            int freq=it.second;
            minu=min(minu,freq);
        }

        int ans=INT_MAX;
        for(int k=minu;k>=1;k--){

            int curr=0;
            bool flag=true;
            for(auto &it:mp){

                int freq=it.second;
                int groups=ceil((double)freq/(k+1));

                //group+1 isliye kyunki pehle k
                //ex: 10 10 10 and groupsize=1
                //to 10 10 and 10 , 3/2=1.5 , o

                if(groups*k>freq){
                    flag=false;
                    break;
                    //maan lo k size ke 4 group bana diye , to elements=4
                    //itne elements present bhi to hone chahiye
                }

                curr+=groups;
            }
            if(flag)
                ans=min(ans,curr);
        }
    }
}
```

